Design of test cases and a maintenance plan for a sample

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Aim

To design a comprehensive suite of test cases and establish a software maintenance plan for an AI Lecture Voice Recorder, Automated Notes Generator, and Important Questions Extraction Application.

Objective

- Identify functional and non-functional requirements for audio recording, speech recognition, summarizing, and question extraction modules.
- Design structured unit, integration, and system test cases for audio processing and natural language processing pipelines.
- Formulate a structured software maintenance plan covering AI model updates, bug fixes, and audio processing performance optimizations.
- Define test execution and maintenance workflows for long-term system stability and transcription accuracy.

Selected Software Application

AI Lecture Recorder & Smart Notes Generator System

Overview

An AI-powered Smart Lecture Assistant allows users to:

- Record live lecture audio in real-time with noise suppression.
- Transcribe speech into structured text using speech-to-text models.
- Automatically generate concise summary notes and key takeaways.
- Extract important exam and study questions based on lecture context.
- Export notes to PDF or Markdown and search past lecture recordings.

Functional Requirements

Audio Recording Module

- Start, pause, resume, and stop high-quality audio recording.
- Background audio recording support while using other apps.
- Audio file compression and secure local or cloud storage.

Transcription & Processing Module

- Real-time or post-recording Speech-to-Text conversion.
- Speaker diarization to distinguish the lecturer from student questions.
- Background noise reduction and audio normalization.

AI Notes & Question Generation Module

- Automatic generation of structured lecture summaries (bullet points, main topics).
- Extraction of key concepts, formulas, and terminology.
- Automated generation of short-answer, long-answer, and multiple-choice questions (MCQs).

User & Library Management Module

- Subject and course folder categorization.
- Search functionality across saved transcripts and generated notes.
- Export notes to PDF, DOCX, and plain text formats.

Design of Test Cases

TC ID	Module	Test Scenario	Input Data	Expected Result
TC_01	Audio Recording	Record Live Audio	Microphone input (60 min lecture audio)	Audio file recorded without audio dropping or file corruption.
TC_02	Transcription	Speech-to-Text Conversion	Recorded lecture WAV/ MP3 file	Accurate text transcript generated with high speech-recognition accuracy.
TC_03	AI Processing	Summary & Notes Generation	Raw transcript text	Structured bullet-point summary and key topics created successfully.
TC_04	AI Processing	Important Questions Extraction	Raw transcript text	Generates relevant study questions (Conceptual, MCQs, Exam prep).
TC_05	System / Export	Export Notes to PDF	Generated notes and questions	PDF file created with proper formatting and a valid download link.

Test Execution & Feedback Cycle

Requirement Analysis ↓ Audio/AI Test Case Design ↓ Automated & Manual Execution ↓ Accuracy & Defect Logging ↓ Model & Patch Deployment ↓ Regression Testing

Maintenance Plan

1. Corrective Maintenance

- Hotfixes for microphone permission issues or audio recording crashes on specific phone models.
- Resolving timeout errors during long audio transcription or API gateway calls.

2. Adaptive Maintenance

- Updating Speech-to-Text and NLP models to improve domain-specific academic vocabulary.
- Updating mobile dependencies to maintain compatibility with new iOS and Android OS releases.

3. Perfective Maintenance

- Optimizing audio compression algorithms to reduce storage usage without degrading transcription quality.
- Enhancing UI performance for instant search across thousands of lecture transcripts.

4. Preventive Maintenance

- Conducting quarterly security and privacy audits to protect student audio recordings and personal data.
- Refactoring core processing pipelines and pruning outdated test datasets.

Maintenance Schedule & Comparison

Maintenance Type	Frequency	Focus Area	Impact on System
Corrective	As needed (Immediate)	Audio crashes & transcription failures	High (Restores immediate app usability)
Adaptive	Quarterly / OS Updates	AI Model tuning & OS compatibility	Medium (Sustains accuracy & platform support)
Perfective	Bi-weekly (Sprint updates)	Compression & UI performance	High (Enhances user experience & speed)
Preventive	Monthly / Quarterly	Data privacy audits & code cleanup	High (Prevents data leaks & system downtime)

Results

- The Lecture Recording and AI Notes Generator System requires dedicated test coverage across audio input, transcription, summary synthesis, and question extraction modules.
- A structured test suite guarantees reliable recording stability, high transcription accuracy, and accurate question generation.
- An active maintenance plan ensures seamless multi-device operation, continuously updated AI models, and strong security for user recordings.

Conclusion

Designing precise test cases combined with a proactive maintenance strategy ensures that AI-driven academic tools remain accurate, performant, and reliable. By systematically testing audio processing workflows and maintaining AI model accuracy, development teams can deliver high-quality automated notes and study materials while maintaining application efficiency and user satisfaction.